

Answer Key — Biology Grade 11

Sub-Strand 2.3: Excretion in Plants • Lesson 2: How Do Plants Get Rid of Wastes? Nature Walk, Exudation and Guttation Evidence

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
C	A	B	B	B	A	C

USE AFTER: PREDICT PHASE — Predict: prediction on whether droplets on arrowroot leaf edges are waste, water or something else

Q1. On an early humid morning, a student says, "The droplets along the edges of the arrowroot leaf are just dew." Which observation would best suggest that the droplets come from the plant itself?

- A. The droplets are clear and have no colour
- B. The droplets are seen only in the morning
- C. The droplets are in the same places along the leaf margins and tips, not spread randomly over the leaf**
- D. The droplets are wet when touched

Correct answer: C. Guttation droplets form at hydathodes, which are fixed spots at leaf margins and tips. Dew settles randomly over the whole surface, so droplets at the same spots suggest the plant is releasing them.

USE AFTER: OBSERVE PHASE — Part A, Nature walk: Station 2, Euphorbia latex from a tiny cut

Q2. A teacher makes a tiny cut on a Euphorbia stem. Thick white milky liquid oozes out and dries to cover the cut. Which waste-removal route does this show?

- A. Exudation of latex through a cut**
- B. Diffusion of gases through lenticels
- C. Transpiration through stomata
- D. Guttation through hydathodes

Correct answer: A. Gums, resins and latex that ooze out through cuts or bark are exudates, so this is exudation. Guttation gives watery droplets at leaf edges, and transpiration loses water vapour.

USE AFTER: OBSERVE PHASE — Part A, Nature walk: Station 4, fallen yellow leaves and leaf scars as evidence of tissue/organ fall

Q3. During the nature walk, learners collect fallen yellow leaves and see leaf scars on a branch. Why can the falling of old leaves count as a way a plant gets rid of wastes?

- A. Leaves fall only because of the season, so wastes are not involved
- B. Wastes stored in the old leaves are removed from the plant when the leaves drop**
- C. Falling leaves reduce transpiration by closing the stomata
- D. Falling leaves release gum and latex through the leaf scar

Correct answer: B. Plants can store wastes in old leaves, bark and fruits. When these parts fall, the stored wastes leave the plant. Seasonal leaf fall is not just a timing event.

USE AFTER: OBSERVE PHASE — Part B, Guttation experiment (POE): bell jar seedling and labelled control seedling

Q4. In the guttation experiment, one well-watered seedling is covered with a bell jar. A second seedling is left uncovered in a drier spot. What is the purpose of the uncovered seedling?

- A. To show that seedlings do not need water
- B. To compare and show whether high humidity makes guttation droplets appear**
- C. To measure how much water the covered seedling absorbs
- D. To make sure the covered seedling gets enough light

Correct answer: B. The control differs only in humidity. Comparing the two seedlings shows whether high humidity causes droplets at the leaf edges.

USE AFTER: EXPLAIN PHASE — Claim-Evidence-Reasoning (CER): explaining the droplets at leaf tips under the jar

Q5. Guttation droplets form at leaf edges on a cool, humid night after the soil has been watered well. Which explanation is correct?

- A. The leaves sweat to cool down, as humans do
- B. Roots absorb water faster than transpiration can remove it in humid air, so liquid water is forced out at hydathodes**
- C. The stomata open wide and release a lot of water vapour
- D. Dew from the air falls and collects on the leaf edges

Correct answer: B. In humid air, transpiration is slow while the roots keep absorbing water. The extra water, with dissolved salts and sugars, is pushed out as liquid at hydathodes.

USE AFTER: EXPLAIN PHASE — Venn diagram contrasting guttation and transpiration; exit ticket 'Contrast guttation and transpiration in 2 sentences'

Q6. Which statement correctly contrasts guttation and transpiration?

- A. Guttation loses liquid water with dissolved substances through hydathodes, mainly at night or early morning. Transpiration loses water vapour through stomata, mainly in the day.**
- B. Guttation releases gum and resin from cuts, while transpiration releases latex.
- C. Guttation loses water vapour through stomata in the day. Transpiration loses liquid water through hydathodes at night.
- D. Guttation and transpiration both lose only water vapour, but guttation happens in the roots.

Correct answer: A. Guttation is liquid water carrying dissolved salts and sugars from hydathodes, mainly in humid conditions at night or early morning. Transpiration is water vapour lost through stomata, mainly in warm, dry daytime conditions.

USE AFTER: DQB PHASE — DQB: sorting new questions into Gaseous, Liquid, Solid/insoluble wastes and Economic importance

Q7. After the nature walk, groups write new questions on the Driving Question Board. Which question is about wastes that are neither gases nor liquids and cannot easily evaporate or ooze out?

- A. Why does Euphorbia latex dry quickly at a cut?
- B. Who buys gum arabic, and for how much?
- C. What happens to wastes that cannot evaporate or ooze out easily?**
- D. Why do guttation droplets appear mostly at night?

Correct answer: C. The unanswered question about insoluble wastes that cannot evaporate or ooze out is the one that leads to the next lesson. The other questions are about guttation, the market value of gum, and latex drying.